Volume V · Chapter 26 — LLMs/VLA as Planners & Human-in-the-Loop Safety · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-V-Chapter-26-context.md` for the AI interaction log.

Prerequisites: Ch 11, Ch 25.

Learning outcomes — the student can: use LLMs/VLA models as high-level planners; translate NL tasks into verified action sequences; design human-in-the-loop approval; describe VLA SOTA and its limits in medicine.

Topics (theory) — 8 h:

#	Topic	h
26.1	LLMs as planners/reasoners for robots	2
26.2	Vision-Language-Action (VLA) models: concepts & SOTA (e.g., RT-2)	2
26.3	Grounding language to actions & verification	2
26.4	Human-in-the-loop: approval gates, oversight, fail-safes	2

Datasets/tools: local LLM; the simulator; a planning framework. **Assessment:** LLM-planner-in-sim with approval gate (**60%**); safety analysis (**20%**); quiz (**20%**). **Key decisions:** autonomy level; verification strategy; **when human approval is mandatory**. **References:** plan §7; §13 → *Robotics & embodied AI*; *Medical LLMs*. **Hours:** Theory **8** + Lab **12** = **20**.